



Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3*, *OsNPF2.4* and *OsAGO1a*, respectively

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ABSTRACT

Rice microRNA168a (*osa-miR168a*) plays important roles in mediating flowering time, grain yield and vigor, seeding growth, and immunity by targeting the RNA-induced silencing complex component Argonaute 1 (*AGO1*). However, the functions of miR168a exerted by targeting other genes require further clarification before it could be used in rice molecular breeding. In this study, we identified a new target gene of *osa-miR168a-5p* (*miR168a-5p*) in rice called *OsOFP3* (*ovate family protein 3*) and investigated the roles of miR168a-5p in response to brassinosteroids (BRs), salt stress, and nitrogen allocation. Up- and downregulated miR168a-5p expression respectively decreased and increased the expression of the BR-negative regulator *OsOFP3*. The results of RNA ligase-mediated rapid amplification of cDNA ends (5'RLM-RACE) revealed cleavage sites in *OsOFP3* and *OsNPF2.4* mRNAs. The phenotype of miR168a-5p transgenic rice was BR-associated and included the lamina bending response to BR, short seeds, and low 1000-grain weight. MicroRNA 168a-5p also regulated the expression of the nitrate transporter, *OsNPF2.4*, which affected nitrogen allocation, and regulated *OsAGO1a* expression in response to salt stress. Taken together, rice miR168a-5p regulates BR-associated pathways, nitrogen transport, and stress by targeting *OsOFP3*, *OsNPF2.4*, and *OsAGO1a*, respectively, resulting in a series of important agronomic traits for rice breeding.

1. Introduction

Highly-conserved, noncoding microRNAs (miRNAs) comprising 20–24 nucleotides (nt) play important roles in plant development and responses to stress (Zhang et al., 2006). Plant miRNAs usually function through the transcriptional cleavage or translational inhibition of target genes, and cleave them via the RNA-induced silencing complex component Argonaute 1 (*AGO1*) protein to degrade a target mRNA (Vaucheret, 2009). MicroRNA168 (*miR168*) is a key miRNA that targets *AGO1* to regulate its expression and play roles in plant growth and environmental stress responses (Zhou et al., 2022).

The miR168 family includes miR168a and miR168b and after a mature miR168a dimer forms, it can be split into miR168a-3p and miR168a-5p (Vaucheret et al., 2004). MicroRNA168a maintains a steady-state balance with *AGO1*. In general, miR168 plays important roles in the physiological and molecular functions of its interaction with

AGO1 (Vaucheret, 2009), and the miR168-*AGO1* module contributes to agronomically important traits in rice (Wang et al., 2021; Zhou et al., 2022). The abscisic acid (ABA) binding factor (ABF) family in Arabidopsis binds to the promoter region of the *MIR168a* gene, resulting in a response to ABA and other stresses (Li et al., 2012). The miR168a-*AGO1* module regulates the growth of the root tip meristem by regulating the miR165/166 pathway (Du et al., 2020).

The *OsAGO1* family in rice comprises four genes *OsAGO1a-d* (Yang et al., 2013). The rice miR168a-*OsAGO1* module regulates tillering, immunity, and flowering time by mediating downstream miR535, miR1320 and miR164, respectively (Wang et al., 2021). Rice miR168a also targets *OsAGO1/18* and plays pleiotropic roles in rice by mediating downstream miR166 d/g-5p, miR167e/i-3p, and miR172d-5p (Zhou et al., 2022). The miR168-*AGO18* module is also involved in the inhibition of rice antiviral RNAi (Wu et al., 2015). To the best of our knowledge, these functions of miR168a are implemented by targeting

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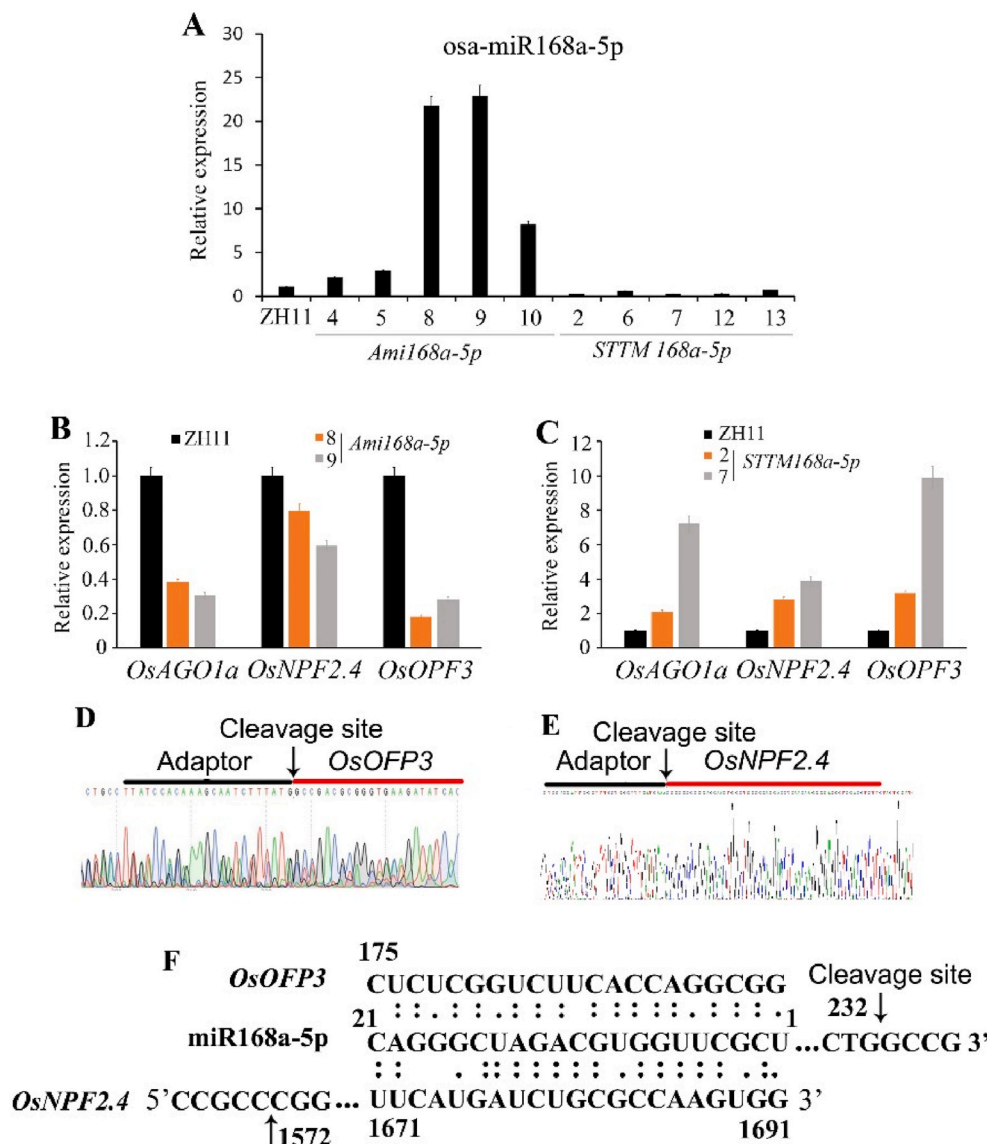


Fig. 1. Identification of target genes of rice miR168a-5p. (A) Relative expression of miR168a-5p in transgenic rice lines with overexpressed miR168-5p (*Ami168a-5p*) and downregulated miR168 expression (*STTM168a-5p*). Relative expression of target genes in *Ami168a-5p* (B) and *STTM168a-5p* (C). One one-month-old rice seedlings were assessed using RT-qPCR. The reference genes for miRNA in (A) and mRNA in (B,C) were *U6* and *OseEF1a* respectively. Results are shown as means $\pm$ SD of three biological repeats. (D–F) Identification of cleavage sites in *OsOPF3* and *OsNPF2.4* mRNA using 5'RLM-RACE. Sequenced chromatogram of *OsOPF3* (D) and *OsNPF2.4* (E), and cleavage sites of *OsOPF3* and *OsNPF2.4* mRNA (F). Arrow indicates cleavage sites on aligned miRNA:mRNA. Results were generated from two biological repeats and 10 sequenced clones.

AGOs in different species.

We predicted that miR168a-5p would target *ovate family protein 3* (*OsOPF3*), a negative brassinosteroid (BRs) synthetic signal regulator (Xiao et al., 2020). Brassinosteroids control cell elongation, division, and differentiation, resulting in specific roles in plant growth and development and responses to stress (Xia et al., 2015). The *OPFs* were initially discovered in tomato (*Solanum lycopersicum*) as regulators of fruit shape (Liu et al., 2002). Overexpressed rice *OPFs* *OsOPF1-3* and *8* lead to growth repression (Schmitz et al., 2015; Xiao et al., 2020; Yang et al., 2016). Rice miR168a represses the expression of *OsNPF2.4*, which regulates seed vigor (Zhou et al., 2020), and miR168 expression responds to nitrogen (Nischal et al., 2012). *OsNPF2.4* belongs to the nitrate and peptide family (NPF) of transporters (Xia et al., 2015); it transports NO_3^- over long distances and indirectly affects potassium recycling between roots and shoots (Xia et al., 2015).

Here, we found that miR168a-5p targets *OsOPF3* (Xiao et al., 2020) that mediates seed length and 1000-grain weight by affecting the BR response, *OsNPF2.4* that mediates nitrogen allocation, and *OsAGO1a* that affects salt tolerance in rice. The present and previous findings (Zhou et al., 2020, 2022; Wang et al., 2021) show that rice miR168a plays several roles in rice development and stress responses.

2. Materials and methods

2.1. Vector construction and rice transformation

We overexpressed miR168a-5p and miR168a-3p by synthesizing a 21-bp sequence and inserting it downstream of the 35S promoter in pCambia1301 (<http://www.cambia.org>), followed by an artificial microRNA technique (<http://wmd3.weigelworld.org/cgi-bin/webapp.cgi>). A short tandem target mimic was used to reduce the expression of miR168a-5p and miR168a-3p (Yan et al., 2012), and *STTM168a-5p* and *STTM168a-3p* constructs, which were amplified from the *STTM1848* vector (Xia et al., 2015) using antisense 21-bp miR168a-5p and 22-bp miR168a-3p primers (Table S1), were inserted downstream of the maize ubiquitin 1 gene (*Ubi-1*) promoter in pXU1301 (Xia et al., 2015). The 35S promoter before *GUS* was replaced with the *OsMIR168a* promoter in pCambia1301 for promoter:*GUS* analysis. These constructs were transformed into japonica rice (*Oryza sativa* L.) var. Zhonghua 11 (ZH11) by *Agrobacterium*-mediated transformation (Hiei et al., 1997).

2.2. Prediction of miR168a-5p mRNA targets

Putative targets of miR168a-5p were identified using online

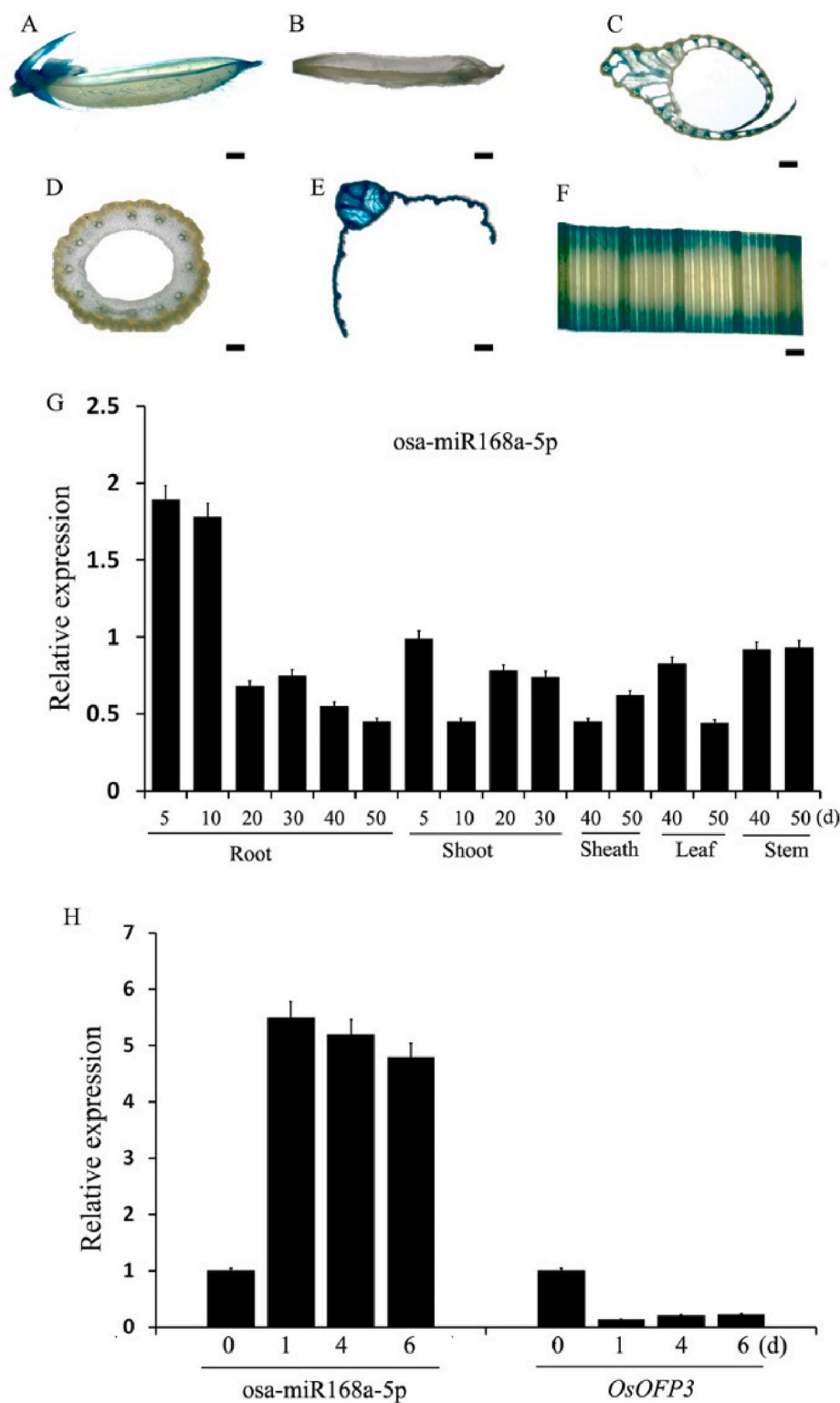


Fig. 2. Expression of miR168a-5p and *OsOPF3*. Beta-glucuronidase (GUS) staining of *Pro_{OsMIR168a}:GUS* transgenic rice (A; bar = 500 μ m) glumes, (B; bar = 500 μ m) anthers, (C) leaf sheaths, (D) stems and (E, F) leaf blades (C–F; bar = 250 μ m). Relative expression of miR168a-5p in organs of ZH11 under normal growth (G), and in leaves treated with BR (H). Plants were grown under normal conditions (A–G), and 10-day-old ZH11 seedlings were grown in Hoagland solution containing 1000 ng/L BR (H). *U6* and *OseEF1a* were reference genes for miRNA (G) and mRNA (H), respectively. Data are shown as means $\pm$ SD of three biological repeats.

prediction programs (miRU: <http://plantgrn.noble.org/psRNATarget/>).

2.3. Total RNA extraction and reverse transcription-quantitative PCR (RT-qPCR)

Total RNA extraction (including miRNA) from rice and RT-qPCR were performed as described (Xia et al., 2012).

2.4. Salt treatment

Rice seedlings were immersed in Hoagland solution (Yoshida et al., 1976) containing 150 mM NaCl to assess gene expression, and sterilized seeds were grown in 1/2 MS medium supplemented with 150 mM NaCl for germination and seedling growth.

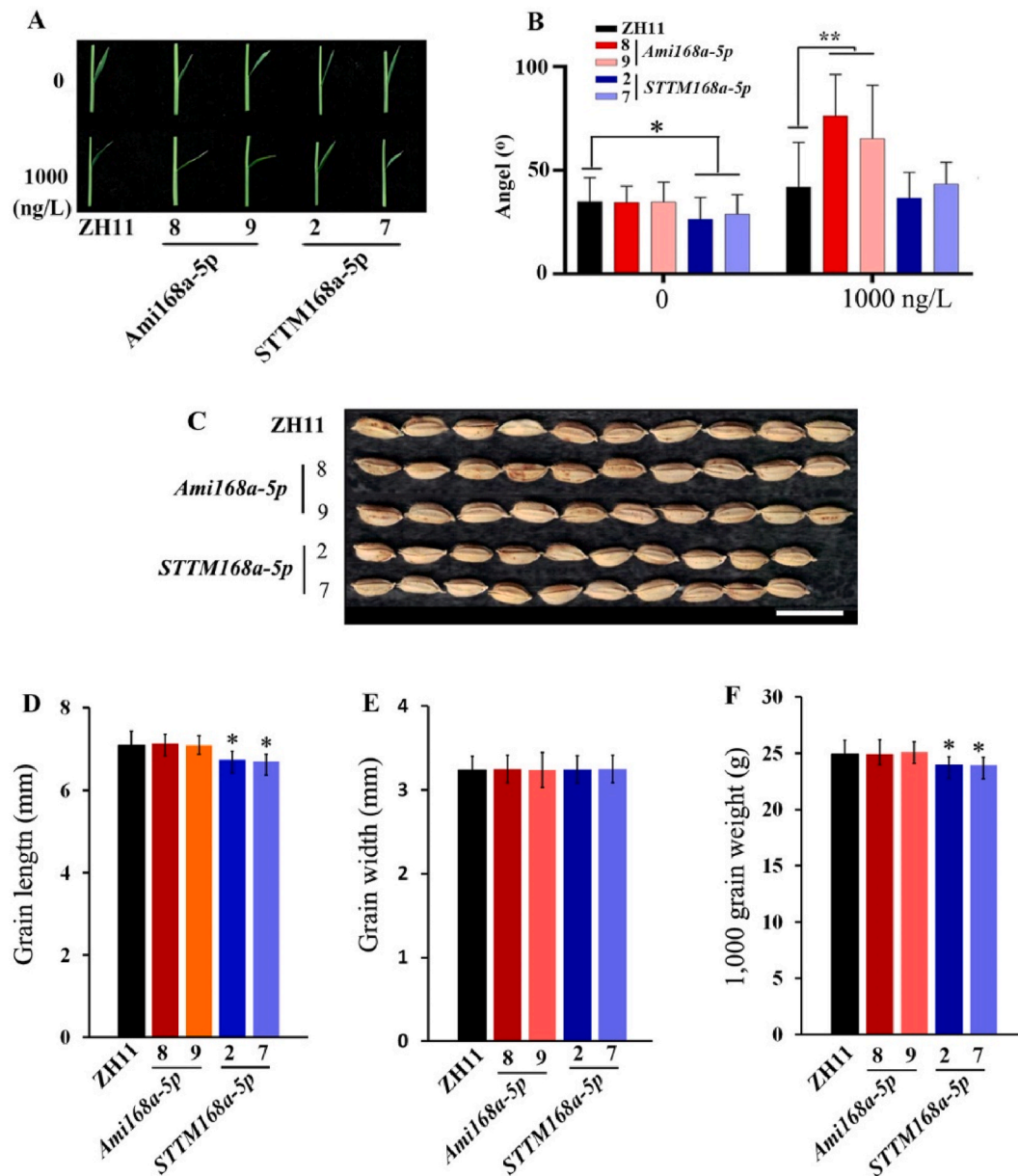


Fig. 3. MicroRNA 168a-5p caused BR-associated phenotypes of transgenic rice. (A) Sensitivity of *Ami168a-5p*, and *STTM168a-5p* transgenic rice lines to BR, compared with wild type (ZH11) determined as Lamina bending and (B) statistical findings. The seedlings were immersed in 1000 ng/L BR for 4 days. Data are shown as means $\pm$ SD of three biological repeats, ** $p < 0.01$, Student t -tests ($n = 34$ plants) (C–F) Seed performance of miR168a-5p transgenic rice. Rice plants grown under normal conditions (C; bar = 0.5 cm) to measure grain length (D), width (E), and 1000 grain (F). Data are shown as means $\pm$ SD of three biological repeats in (D–F). * $p < 0.05$ Student t -test ($n > 200$ seeds). BR, brassinosteroid.

2.5. Nitrogen treatment

For gene expression, rice seedlings were placed in Hoagland solution base (Yoshida et al., 1976) with low (LN), normal (NN) and high (HN) nitrogen concentrations (0.75, 3.0, and 6.0 mM NH_4NO_3 , respectively).

2.6. Brassinolide (BL) treatment

We dissolved plant hormone 24-epibrassinolide (BL; Sigma-Aldrich Corp., St. Louis, MO, USA) in 100% methanol and added it to Hoagland solution (Yoshida et al., 1976). The control was Hoagland solution containing methanol without BL. Lamina joints were analyzed in second-leaf laminae of 4-week-old seedlings as described (Yang et al., 2018).

2.7. Beta glucuronidase (GUS) histochemical staining

GUS histochemical staining was performed as previous description (Jefferson et al., 1987). Samples were examined using a DVM6 light microscope (Leica, Microsystems GmbH, Wetzlar, Germany).

2.8. RNA ligase-mediated rapid amplification of cDNA ends (5'RLM-RACE)

Total RNA from tillering stage leaf sheaths was directly ligated to GCUGAUGGCGAUGAAUGAACACUGCGUUUGCUGGCUUGAUGAAA, a synthesized RNA adaptor that has a 5'-hydroxyl group at both ends and can only ligate 5'-phosphorylated RNAs, including the truncated products of microRNA-guided mRNA cleavage. The ligated product was directly reverse-transcribed using an oligo(dT) primer. The cDNA was

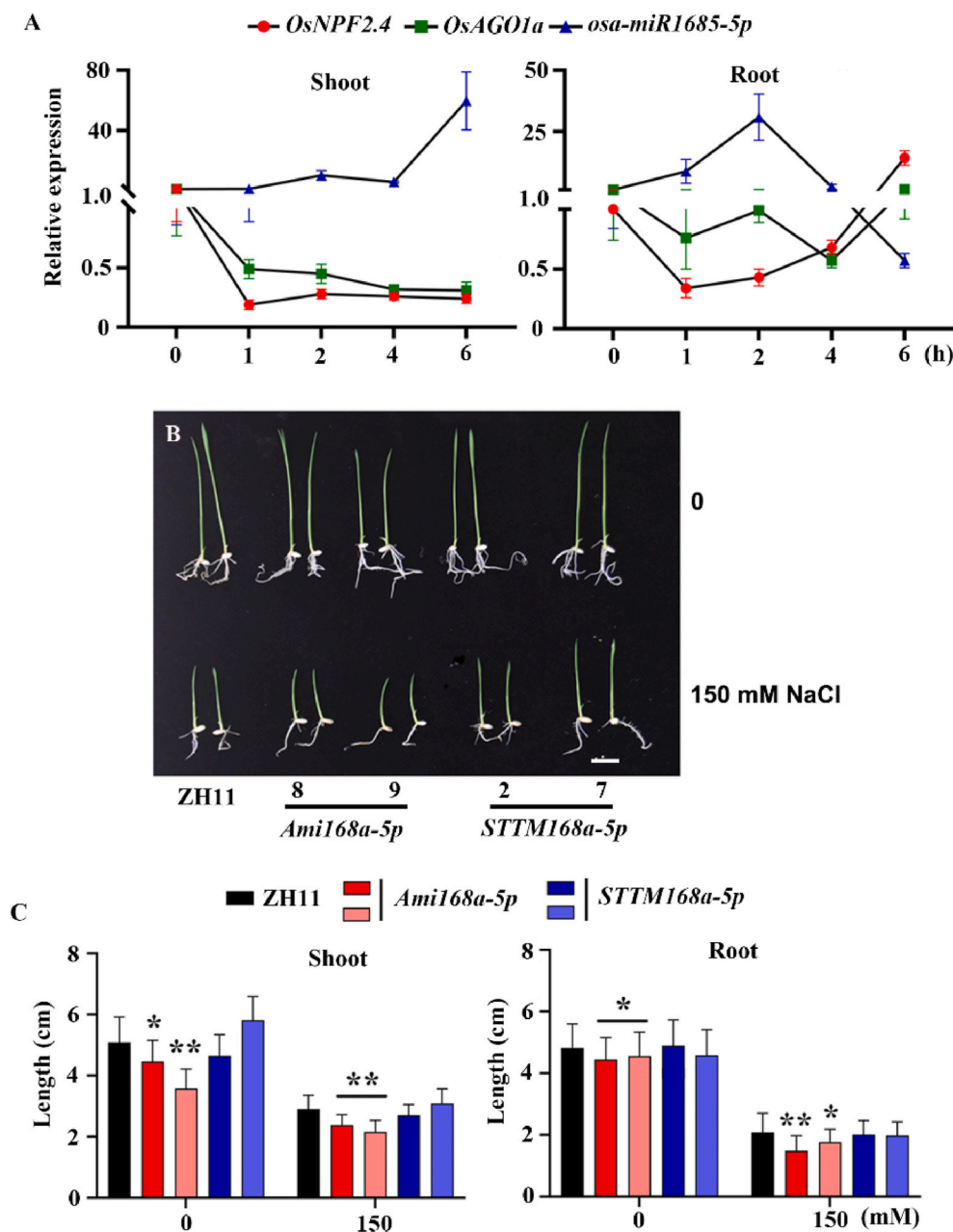


Fig. 4. MicroRNA 168a-5p mediates salt tolerance of rice. (A) Altered expression of miR168a-5p and its target genes *OsNPF2.4* and *OsAGO1a* in ZH11 seedlings immersed in 150 mM NaCl. Reference genes for miRNA and mRNA were *U6* and *OseEF1a*, respectively. Data are shown as means $\pm$ SD of three biological repeats. Growth performance (B) and statistical data (C) of *Ami168a-5p*, *STTM168a-5p* and ZH11 seedlings immersed in 1/2 MS medium without or with 150 mM NaCl for 7 days at 25 °C. Bar = 1 CM. Data are shown as means $\pm$ SD of three biological repeats, * $p < 0.05$, ** $p < 0.01$, Student *t*-test ($n = 80$ seedlings).

amplified by nested PCR, and the final PCR product was gel-purified and subcloned into the pGEM-T Easy Vector (Promega Corp., Madison, WI, USA) for sequencing.

2.9. Measurement of total carbon and nitrogen

Whole, mature rice plants were selected and cleaned with tap and purified water. The roots, stems, leaves, leaf sheaths, and panicles were oven-dried at 100 °C for 0.5 h then held at 70 °C until all samples reached a constant weight. The dried samples were crushed in a LUCKM-24 mill prototype (Guangzhou Luka Corp, Guangzhou, China) and passed through a 100-mesh sieve. Total nitrogen and carbon contents were determined by stable isotope analysis using an IsoPrime 100 stable isotope mass spectrometer (Elementar Analysensysteme GmbH, Langensfeld, Germany).

3. Results

3.1. *OsOFP3* is a target of miR168a-5p

We predicted that *OsOFP3* would be a target gene of *osa-miR168a-5p* (miR168a-5p) using psRNAtarget (<http://plantgrn.noble.org/psRNAtarget/home>). We constructed transgenic rice lines overexpressing (*Ami168a-5p*) and downregulated (*STTM168a-5p*) miR168a-5p to verify our prediction (Fig. 1A). The RT-qPCR results showed that upregulated and downregulated miR168a-5p expression respectively downregulated and upregulated *OsOFP3* expression in *Ami168a-5p* and *STTM168a-5p* (Fig. 1B and C), in the same way as the two established targets (*OsNPF2.4*, and *OsAGO1a*) (Nischal et al., 2012; Zhou et al., 2020, 2022; Wang et al., 2021). We further selected *japonica* rice Zhonghua 11 (ZH11) and *Ami168a-5p* to identify cleavage sites of *OsOFP3* mRNA using 5'RLM-RACE (Fig. 1D, F). The sequencing results showed that miR168a-5p directs the cleavage site of *OsOFP3* mRNA 56 nt

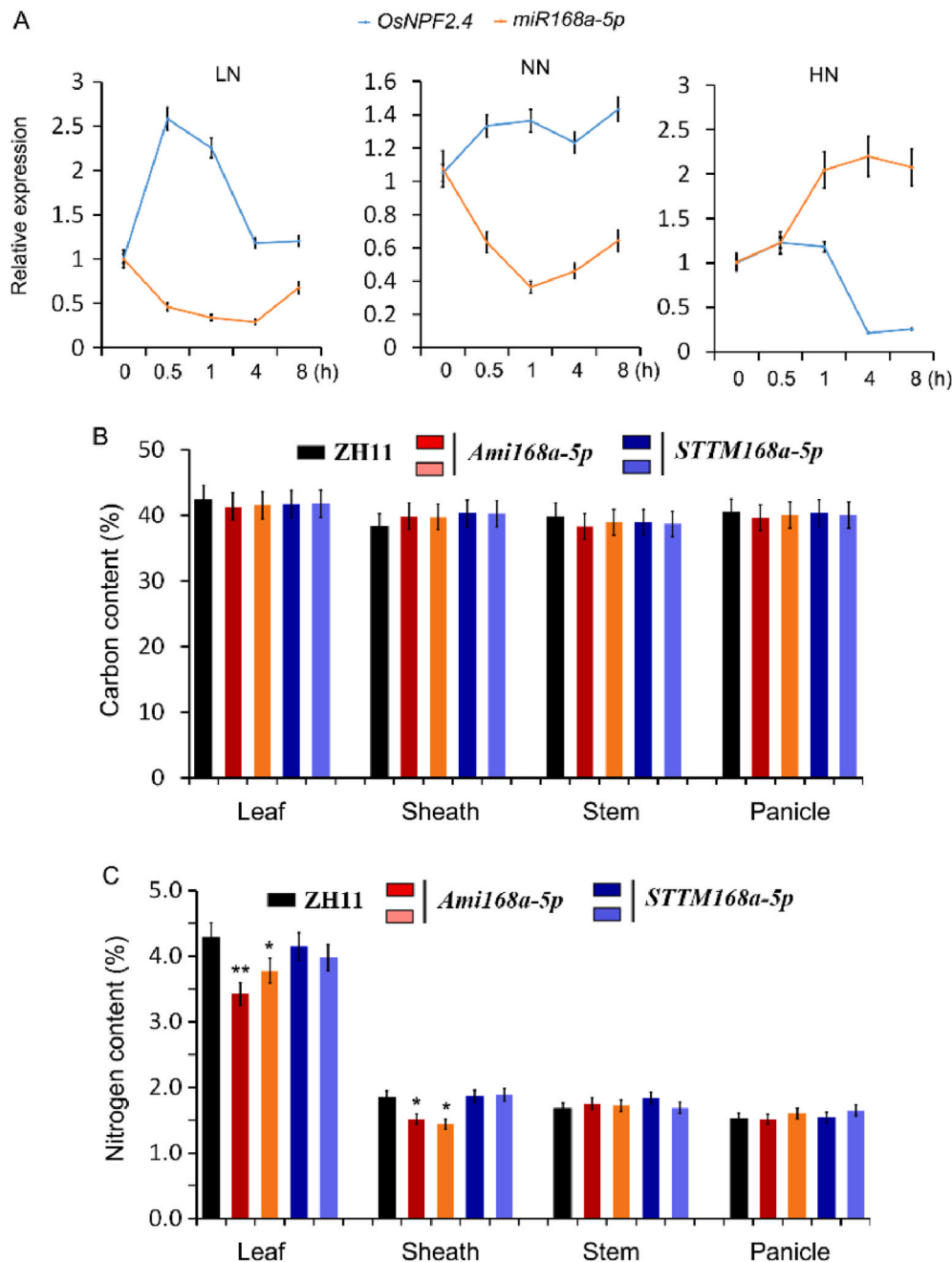


Fig. 5. MicroRNA 168a-5p responds to nitrogen. (A) Expression of miR168a-5p and *OsNPF2.4* in ZH11 under 0.75 (LN), 3.0 (NN) and 6 (HN) mM NH_4NO_3 . Reference genes for miR168a-5p and *OsNPF2.4* were *U6* and *OseEF1a*, respectively. Data are shown as means $\pm$ SD of three biological repeats. Total carbon (B) and nitrogen (C) ratios (%) in organs of ZH11, *Ami168a-5p*, and *STTM168a-5p* plants grown under normal conditions in a controlled field.

downstream of the complementary site of miR168a-5p/*OsOFP3* mRNA. These results indicated that *OsOFP3* is a target gene of miR168a-5p.

3.2. Expression of *OsOFP3* and miR168a-5p inversely changed in response to BR

We investigated the role of miR168a-5p by constructing transgenic rice with the *GUS* gene driven by the *OsMIR168* promoter (Fig. 2A–F). The results of staining Pro_{*OsMIR168*}:*GUS* rice with GUS showed that miR168a-5p is expressed in rice glume, anthers, leaves sheaths, and stems. Temporal and spatial expression of miR168a-5p in ZH11 determined by RT-qPCR also showed that miR168a-5p is expressed at almost all stages of rice growth and development and is abundant in the roots of seedlings (Fig. 2G). Because *OsOFP3* plays a distinctly negative role in

the BR response (Xiao et al., 2020), we investigated whether the expression of miR168a-5p and *OsOFP3* inversely changes in response to BR (Fig. 2H). The RT-qPCR data showed that BR respectively upregulated and downregulated the expression of miR168a-5p and *OsOFP3*. These results showed that the change in *OsOFP3* expression is opposite that of miR168a-5p in response to BR.

3.3. MicroRNA 168a-5p caused BR-related phenotypic changes in transgenic rice

We analyzed the BR-related phenotypes of miR168a-5p transgenic rice to determine whether miR168a-5p functions in the response to BR by mediating *OsOFP3* expression (Fig. 3A and B). Rice lines showing the highest (*Aim168-5p-2/7*) and lowest (*STTM168-5p-8/9*) miR168a-5p

expression (Fig. 1A) were analyzed. We assessed the sensitivity of *Ami168a-5p* and *STTM168a-5p* to BR as leaf angle bending (Fig. 3A and B). The leaf angle of *Ami168a-5p* did not significantly differ from that of ZH11, whereas that of *STTM168a-5p* was smaller in the absence of BL. The leaf angle of *Ami168a-5p* was significantly greater than that of ZH11 and *STTM168a-5p* in response to 1 µg/L BL, but the difference in the leaf angle between them disappeared. These results indicated that miR168a-5p changed the response of rice leaf angles to BR. The main role of BRs in regulating cell elongation is to regulate plant height and seed length (Yang et al., 2018). Therefore, we investigated the role of miR168a-5p in rice seed development. The results showed that *STTM168a-5p* grains were shorter, and the 1000-grain weight was lower than that of ZH11 (Fig. 3C–F). Overexpressed and downregulated miR168a-5p caused similar BR-related phenotypic changes to those of *OsOPF3* (Xiao et al., 2020). However, the plant height and tiller number of miR168a-5p transgenic rice did not differ from those of ZH11 (Fig. S1). In addition, neither overexpressed nor downregulated miR168a-3p elicited these phenotypic changes (Fig. S2). These results suggested that miR168a-5p functions in rice by regulating *OsOPF3* expression.

3.4. MicroRNA 168a-5p mediates salt response of rice

Plant miR168a functions in responses to biotic stress in rice (Wang et al., 2021), and in tolerance to abiotic stresses (Li et al., 2012) by mediating the expression of *AGO1*. Altered expression of miR168a-5p also affected *OsAGO1a* expression (Fig. 1B and C) like that of full-length miR168a (Zhou et al., 2022). Therefore, we investigated whether miR168a-5p is also involved in the response to salt stress in rice. The expression of miR168a-5p was induced by NaCl in seedling shoots and roots, but that of its target genes *OsAGO1a* and *OsNPF2.4* was reduced (Fig. 4A). At the germination stage, *Ami168a-5p* seedlings grew more slowly than those of ZH11 (Fig. 4B and C), but the height of mature plants did not differ among ZH11, *Ami168a-5p*, and *STTM168a-5p* (Fig. S1), which was unlike that of miR168 overexpression (Wang et al., 2021) and *MIR168a*-mutation (Zhou et al., 2022). However, NaCl inhibited the seedling growth of *Ami168-5p* more than that of ZH11 (Fig. 4B and C), suggesting that overexpressed miR168a-5p decreases the germination rate and salt tolerance of seedlings. These findings indicated that miR168a-5p is involved in the responses of rice to salt.

3.5. MicroRNA 168a-5p is involved in nitrogen response of rice

Rice miR168a represses expression of the nitrate transporter *OsNPF2.4/OSPTR2* (Xia et al., 2015; Zhou et al., 2020). Our results also showed that the expression of *OsNPF2.4* was repressed by overexpressing miR168a-5p in *Ami168a-5p* (Fig. 1B) and promoted by downregulated miR168a-5p in *STTM168a-5p* (Fig. 1C). We used 5'RLM-RACE to identify cleavage sites of *OsNPF2.4* mRNA to verify whether it is indeed a target gene of miR168a-5p (Fig. 1E and F). Sequencing results showed that miR168a-5p directs the cleavage site of *OsNPF2.4* mRNA 101 nt upstream of the miR168a-5p/*OsNPF2.4* mRNA complementary site, confirming that *OsNPF2.4* is a target gene of miR168a-5p.

We investigated whether miR168a-5p would affect the responses of rice to three concentrations of nitrogen by mediating the levels of *OsNPF2.4*, a pH-dependent, low-affinity nitrate transporter (Xia et al., 2015) (Fig. 5A). The results showed that 0.75 and 3.0 mM NH_4NO_3 , respectively repressed and induced miR168a-5p and *OsNPF2.4*, whereas 6.0 mM NH_4NO_3 respectively induced and repressed miR168a-5p and *OsNPF2.4* expression. These changes in *OsNPF2.4* expression in response to nitrogen are similar to that previous found (Xia et al., 2015). We then measured the changes in carbon and nitrogen contents of *Ami168a-5p* and *STTM168a-5p* (Fig. 5B and C). The total carbon content of *Ami168a-5p* did not significantly differ, but the total nitrogen content in leaves and leaf sheaths was significantly lower than that of ZH11. Therefore, we speculated that miR168a-5p also regulates nitrogen

allocation in rice by targeting *OsNPF2.4*.

4. Discussion

The miR168a-*AGO1* regulatory module was first reported in Arabidopsis, and it participates in abiotic and disease resistance and development in rice (Vaucheret, 2009; Wang et al., 2021; Xian et al., 2014; Zhou et al., 2022). Tomato *MIR168* influences phase transition, leaf epinasty, and fruit development via *SLAGO1s* (Xian et al., 2014). Arabidopsis miR168a targets *AGO1* to mediate its response to ABA and salt (Li et al., 2012). Rice miR168a also influenced resistance to *Magnaporthe oryzae*, grain yield, and flowering time by targeting *OsAGO1* (Wang et al., 2021), and other phenotypes by targeting *OsAGO1a/b/d* and *OsAGO18* (Zhou et al., 2022). Here, we identified a new target gene (*OsOPF3*) of rice microRNA 168a-5p, and roles of microRNA 168a-5p in BR response, salt tolerance, and nitrogen allocation.

The miR168 family in rice includes miR168a and miR168b (Vaucheret, 2009; Zhou et al., 2020). Mature miR168a sequences consists of miR168a-5p and miR168a-3p. Previous findings of rice miR168a were derived from the combined results of these mature sequences (Wang et al., 2021; Zhou et al., 2022). Here, we investigated interactions between miR168a-5p and its target genes. Expression of microRNA168a-5p and *OsAGO1a* in transgenic rice *Ami168a-5p* was upregulated and downregulated, respectively (Fig. 1A and B), and expression of microRNA168a-5p and *OsAGO1a* in transgenic rice *STTM168a-5p* was downregulated and upregulated, respectively (Fig. 1A, C). These showed that rice microRNA168a-5p also mediates *OsAGO1a* expression as that of microRNA 168a (Wang et al., 2021), which then affects rice tolerance of salt stress (Fig. 4) like Arabidopsis (Li et al., 2012). However, overexpressed or downregulated miR168a-5p generated phenotypes other than that of miR168 suppression (Wang et al., 2021) and *OsMIR168a* mutation (Zhou et al., 2022), such as seedling growth at the seed germination stage (Fig. 4C) and plant height at the mature stage (Fig. S1). Overexpressed and downregulated miR168a-3p did not result in phenotypic changes under normal conditions (Fig. S2). Therefore, we speculated that the altered expression of miR168a-5p has similar and different functions from that of full-length miR168a in rice (Wang et al., 2021; Zhou et al., 2022).

We determined that *OsOPF3* as a novel target gene of miR168a-5p (Fig. 1). Changes in *OsOPF3* expression were the opposite of those of miR168a-5p in transgenic rice plants (Fig. 1A–C), and we confirmed that miR168a-5p directed the cleavage sites of *OsOPF3* mRNA (Fig. 1D, F). Brassinolide also induced opposite expression of them (Fig. 2H). These results indicated that miR168a-5p targets *OsOPF3* mRNA and mediates its expression. As an inhibitor of BR synthesis and signaling pathways, *OsOPF3* affects BR that mediates seed development and other processes (Xiao et al., 2020). MicroRNA 168a-5p also mediated changes in leaf angles (Fig. 3A and B). Overexpressed miR168a-5p increased sensitivity to BR, whereas the transgenic rice lines *STTM168a-5p* and *Ami168a-5p* had increased and decreased sensitivity to BR, respectively (Fig. 3A and B). Seeds were short and the 1000-grain weight was light in *STTM168a-5p* (Fig. 3C–F). These phenotypic changes caused by miR168a-5p in transgenic rice are consistent with changes in *OsOPF3* (Xiao et al., 2020). *Ovate family protein 3* belongs the *OPF* gene family and plays important roles in regulating plant growth and development, especially fruit development (Liu et al., 2002, 2014). These results further confirmed that *OsOPF3* is the target gene of miR168a-5p, which mediates rice seed development.

MicroRNA 168a responds to salt, drought and ABA stress by mediating *AGO1* expression in Arabidopsis (Li et al., 2012). We verified that miR168a-5p elicits similar responses in rice. Altered expression of miR168a-5p changed that of *OsAGO1a* (Fig. 1A–C), and the response to salt (Fig. 4A). These results suggested that miR168a-5p regulates *OsAGO1a* expression and mediates the response of rice to salt stress (Fig. 4B and C). MicroRNA 168a-5p represses the expression of *OsNPF2.4* that regulates seed vigor (Zhou et al., 2020). The rice *MIR168a* gene is also

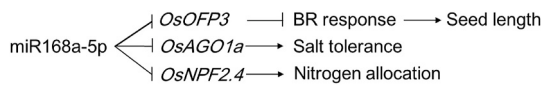


Fig. 6. Working model of rice microRNA 168a-5p. In rice, brassinosteroid induces miR168a-5p expression and represses *OsNPF2.4* expression to mediate seed length, low and normal nitrogen downregulates miR168a-5p expression and upregulates *OsNPF2.4* to mediate nitrogen allocation, NaCl induces miR168a-5p expression and represses *OsAGO1a* expression to affect salt tolerance. ⊥, repress; → promote.

differentially expressed after exposure to low nitrogen (Nischal et al., 2012). We confirmed that miR168a-5p targeted and regulated the expression of *OsNPF2.4* mRNA by identification of the digestion site (Fig. 1E and F), and response to nitrogen (Fig. 5A). The expression of miR168a-5p was suppressed under low and normal nitrogen concentrations, resulting in increased *OsNPF2.4* expression. However, miR168a-5p expression was depressed by high nitrogen, resulting in increased *OsNPF2.4* expression (Fig. 5A). These results confirmed that miR168a-5p regulates the expression of *OsNPF2.4* under different levels of nitrogen. *OsNPF2.4* is a nitrate transporter that affects nitrogen allocation (Xia et al., 2015). Total carbon contents did not significantly differ between *Ami168a-5p* and ZH11 (Fig. 5B), but the total nitrogen content in the leaves and leaf sheaths of *Ami168a-5p* was significantly decreased (Fig. 5C). This indicated that miR168a-5p in *Ami168a-5p* affects nitrogen transport *in vivo* by targeting *OsNPF2.4*. Rice miR168a-5p can affect nitrogen allocation within various rice organs by targeting *OsNPF2.4*. However, the miR168a-*OsAGO1* regulatory pathway can regulate many target downstream miRNAs, resulting in the mediation of rice flowering time, grain yield, and tillering (Wang et al., 2021; Zhou et al., 2022). Therefore, it is difficult to accurately distinguish whether the trait changes are caused by the miR168a-5p/other target genes module or by the miR168a-AGO1-targeting miRNAs module.

Taken together with published information about miR168a (Wang et al., 2021; Zhou et al., 2020, 2022), we proposed a working model of miR168a-5p, which targets *OsAGO1a*, *OsNPF2.4*, and *OsNPF3*, resulting in the regulation of rice salt tolerance, nitrogen allocation, and seed length, respectively (Fig. 6). These findings might lead to more accurate utilization in the molecular breeding of rice *MIR168a*.

CRediT authorship contribution statement

Kuaifei Xia: Investigation. Mingyong Zhang: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jplph.2022.153905>.

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